

Geometric Interpretation of Beam-Link Dynamics

1. Effective Curvature Definition

In the study of beam-link dynamics, we define an effective curvature K based on relative acceleration a and separation distance D . This measure provides a geometric interpretation of the divergence of trajectories.

The basic definition is:

$$K = a / D$$

This implies that large acceleration over short distances signifies high curvature, whereas small acceleration over large distances suggests a nearly flat geometry.

2. Dimensionless Formulation

Incorporating relative velocity v allows for a dimensionless curvature measure:

$$K = (a \times D) / v^2$$

This formulation is consistent with the classical radius of curvature R for circular motion, where $a = v^2 / R$, yielding $K = 1 / R$.

3. Spacetime Coherence Curvature

For the proposed laser-bolas system, we define the **Spacetime Coherence Curvature** (K_c) which accounts for the time-evolution of the separation distance:

$$K_c = a_r / (D_0 + v_r t + 1/2 a_r t^2)$$

The resulting coherence C is defined as:

$$C = 1 / (1 + K_c)$$

As the effective geometric curvature K_c increases due to higher relative acceleration, the coherence of the beam-link dynamics is reduced.